Field Crops Research 300 (2023) 108991

Contents lists available at ScienceDirect

Field Crops Research

journal homepage: www.elsevier.com/locate/fcr

Harvest index has increased over the last 50 years of maize breeding

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ARTICLE INFO

Keywords:
Genetic gain
Maize biomass
Plant breeding
Soil carbon
Stover

ABSTRACT

Context: Quantifying historical changes in maize harvest index (HI), the fraction of above-ground biomass allocated to grain yield, can enhance our ability to explain grain yield trends and estimate stover carbon inputs for sustainability assessments. However, the HI genetic gain has not been the primary focus of previous era studies.

Objective: The aim of this study is to enhance our knowledge of maize HI genetic gain. Our first objective is to quantify HI genetic gain in Bayer Crop Science Legacy hybrids and investigate the contribution of breeding and agronomic management. Our second objective is to develop a general-use model to describe the temporal evolution of maize HI.

Methods: We studied 54 commercial hybrids (103-day and 111-day relative maturities) released from 1983 to 2020 across 13 environments, including plant density (current and historical increasing rate) and N-fertilizer (low and sufficient N rates) treatments. The HI was estimated at physiological maturity by destructively sampling plants. Then we synthesize new experimental data with literature findings (n = 16) to provide a robust HI genetic gain estimate.

Results: Results showed that HI has increased over the years from 0.516 to 0.571 in 103-day hybrids and from 0.537 to 0.584 in 111-day hybrids. The genetic gains were similar across environments and management treatments within the studied range, indicating that this increase is attributed to maize breeding. The N-fertilizer treatments affected the magnitude of the HI, but plant density did not. Our results, combined with 16 literature datasets, revealed a 0.26% year^{-1} relative increase in HI since 1964. We estimated that the increase in HI accounts for ca. 15% of the historical maize yield increase in the US Corn Belt over the past 50 years.

Conclusions: The maize HI has increased over the last 50 years, and this increase was attributed to breeding, not to management.

Significance: Our findings enhance our knowledge of maize HI, will support robust estimations of carbon inputs in sustainability studies, and inform crop models to better capture historical yield increases.

1. Introduction

Maize (*Zea mays* L.) breeding programs have historically targeted grain yields, but this singular focus has indirectly altered other secondary plant traits (DeBruin et al., 2017; Mueller et al., 2019). One such trait is the grain harvest index (HI), the fraction of grain yield to the above-ground biomass (Donald, 1962; Hay, 1995). This trait is routinely used in many disciplines, including crop physiology to evaluate reproductive efficiency (Haegele et al., 2013), soil science to estimate stover carbon inputs (Johnson et al., 2006), nutrient budgets (Ludemann et al.,

Abbreviations: HI, harvest index; N, nitrogen; G, genotype; E, environment; M, management.

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https://doi.org/10.1016/j.fcr.2023.108991
Received 8 February 2023; Received in revised form 24 May 2023; Accepted 1 June 2023
Available online 7 June 2023

2022), and agricultural systems modeling (Neitsch et al., 2011). Therefore, enhancing our knowledge and capacity to predict HI becomes critically important.

Maize hybrids, management practices, and environmental conditions can affect the HI (Hutsch and Schubert, 2017). DeLougherty and Crookston (1979) and Echarte and Andrade (2003) reported variability in HI across maize hybrids. Recent work by Gonzalez et al. (2018) confirmed this variability in a group of genotypes with contrasting density tolerance and yield potential. Vega et al. (2000) and Echarte and Andrade (2003) revealed that the HI is relatively stable for mid-sized plants, being slightly reduced when individual plants are large and sharply reduced in very small plants. Therefore, HI can be affected by factors influencing plant growth, such as plant density (Vega et al., 2000; Echarte and Andrade, 2003), N-fertilization (Ciampitti and Vyn, 2011; Chen et al., 2016), or planting date (Bonelli et al., 2016). Despite this, literature studies still use a constant HI in their evaluations, which might result in misestimations (Prince et al., 2001; Huang et al., 2007; Lee and Tollenaar, 2007; Messina et al., 2022).

Studies on historical changes in HI due to maize breeding are limited, and the few available examined a small number of genotypes and environments. Studies in the 1980s showed little or no increase in HI between 1920 and 1980 (Tapper, 1983; Meghji et al., 1984; Russell, 1985; Tollenaar, 1989), reporting average values of 0.48–0.51. Open-pollinated genotypes used in the US up to the 1920s already had values of around 0.45 (Russell, 1991), and the slight increase between 1920 and the 1980s was explained by increased barrenness of older hybrids under high plant densities and not by greater HI of the newer hybrids (Duvick, 2005). In the last half-century, studies in the US Corn Belt reported positive genetic gains for HI, though their estimations were limited to a few hybrids and environments (Haegele et al., 2013; Chen et al., 2016; Mueller et al., 2019).

Despite the importance of HI for productivity and sustainability assessments (i.e., carbon inputs, nutrient budgets), the genetic gain of HI has not been the primary focus of previous era studies. The lack of robust knowledge of changes in secondary traits, such as HI, may lead to misinterpretations of the relative contribution of genetics, management, and environmental factors affecting the historical maize yield increase. Here we aim to increase our knowledge base on maize HI. Our specific objectives are:

1) determine maize HI genetic gain for Bayer Crop Science Legacy hybrids (54 hybrids belonging to 103-day and 111-day relative maturity) and dissect the contribution of maize breeding from agronomic management (plant density and N-fertilizer)
2) synthesize our findings with previously published articles to develop a general-use model to describe the temporal evolution of maize HI.

We hypothesize that the HI has increased over the years, and that this increase is due to breeding, not to agronomic management.

2. Material and methods

2.1. Experimental locations, designs, and genotypes

Thirteen on-farm experiments were conducted across the US Corn Belt in 2021 (Fig. 1, Table 1). We studied 54 Bayer Crop Science Legacy hybrids commercialized from 1983 to 2020, representing different brands (Table 2). Hybrids were selected based on their commercial relevance for each quinquennium and were all conventional with no GMO traits. The 13 experiments were grouped into four categories (Table 1):

- Four G×E (genotype by environment) experiments with twenty-four 103-day hybrids. The experimental design was a randomized complete block with two replications. The plot size was 15 m^2 (4-row plot by 5 m).

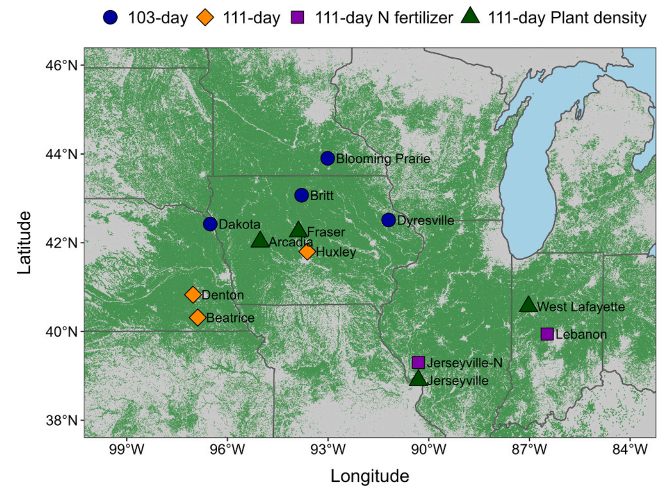

Fig. 1. : Geographic distribution of the field experiments across the US Corn Belt. Different symbols refer to different types of experiments (see legend and Table 1). The shaded green area represents the maize cropland area from 2019 and 2020 (USDA-NASS, 2022a).

- Three G×E experiments with twenty-three 111-day hybrids. The experimental design was a randomized complete block with two replications. The plot size was 15 m^2 (4-row plot by 5 m).
- Two G×E×M (N-fertilizer) experiments with twenty-three 111-day hybrids. The nitrogen (N) fertilizer treatments evaluated were a low (45 kg N ha^{-1}) and a sufficient N rate (204 kg N ha^{-1}). The experimental design was a randomized split-plot complete block design with two replications. N levels were the main plot factor, and hybrids were the split-plot factor. The plot size was 15 m^2 (4-row plot by 5 m).
- Four G×E×M (plant density) experiments with eighteen 111-day hybrids. The plant densities consisted of the current (8.1 plants m^{-2}) and a historical increasing plant density rate: 4.7, 5.9, 7.0, and 8.1 plants m^{-2} for hybrids released in eras 1985, 1995, 2005, and 2015, respectively. These values reflect the historical changes in plant density in the US Corn Belt (USDA-NASS, 2022b). The experimental design was a randomized complete block design with three replications. The plot size was 60 m^2 (8-row plot by 10 m).

2.2. Crop management

Crop planting dates, plant density, and N-fertilizer rates for each experiment followed common regional practices, except for the N-fertilizer and plant density experiments in which N rates and plant densities were experimental treatments, respectively (Table 1). Maize hybrids were sown with precision planters used in Bayer's breeding program, and plant counts indicated no significant hybrid × density interaction in the final plant counts. The planting date ranged from April 15 to May 6, and the final plant population ranged from 6.5 to 8.5 plants m^{-2} across locations, except for the plant density experiments that ranged from 4.7 to 8.4 plants m^{-2} (Table 1). Row spacing was 76 cm in all experiments. Tillage operations, fertilizer amounts, and weed and insect management were decided independently by each cooperating farmer. No weed pressure was observed during reproductive growth at any location. The applied N fertilizer rate ranged from 45 to 274 kg N ha^{-1}. The previous crop was soybean [*Glycine max* (L.) Merr.] in all locations, except for Dyersville (IA), where the previous crop was maize. All locations were rainfed except for Beatrice (NE), which received an additional 89 mm from irrigation.

2.3. Soil and weather conditions

The experimental locations captured major maize production regions

Table 1
Soil properties (S, silty; C, clay; L, loam; OM, organic matter; water table influence), management practices, and average grain yield across experiments. All locations were in a maize-soybean sequence, except for Dyersville (IA), which was in continuous maize.

Experiment	Location	Soil properties			Management practices				Grain yield
		Texture	OM	Water table	Planting date	Plant density	N rate	Tillage system	
			%			plants m^{-2}	kg N ha^{-1}		Mg ha^{-1}
103-day	Britt, IA	CL	3.5	Yes	May 1	7.9	151	Minimum	13.2
	Dyersville, IA	CL	4.1	Yes	April 28	7.5	146	Minimum	11.7
	Dakota, NE	SC	2.3	Yes	April 29	8.3	274	Minimum	13.5
	Blooming Prairie, MN	SC	4.0	Yes	May 5	6.6	250	Convent.	14.2
111-day	Denton, NE	SCL	2.2	No	April 26	6.5	176	No till	11.7
	Beatrice, NE	SC	2.1	No	April 26	8.5	238	No till	15.2
	Huxley, IA	CL	2.8	Yes	May 4	7.9	224	Minimum	14.1
111-day N-fertilizer	Jerseyville, IL	SCL	2.3	Yes	April 15	7.5	45	Convent.	9.8
							202		14.5
	Lebanon, IN	SCL	3.5	Yes	May 6	7.5	45	Minimum	13.0
							205		15.9
111-day Plant density	Arcadia, IA	SC	3.1	No	May 6	8.3	213	Minimum	14.1
						Hist*			13.8
	Fraser, IA	CL	3.3	Yes	April 30	8.4	202	Convent.	14.9
						Hist			15.0
	Jerseyville, IL	SCL	2.2	Yes	April 15	8.0	202	Convent.	14.1
						Hist			13.6
	W Lafayette, IN	CL	2.8	Yes	May 2	7.8	224	Convent.	14.1
						Hist			13.8

* Historical plant density was 4.7, 5.9, 7.0, and 8.1 (7.8–8.4) plants m^{-2} for hybrids released in eras 1985, 1995, 2005, and 2015, respectively.

Table 2
Year of hybrid release (YOR), name, and relative maturity (RM) used in the experiments described in Table 1. All hybrids were conventional with no GMO traits.

ERA	103-day hybrids				111-day hybrids					
	Name	RM	YOR	G×E	Name	RM	YOR	G×E	G×E×M N-fertilizer	G×E×M Plant density
1985					T1100	110	1983	✓	✓	✓
	T1000	100	1983	✓	LH119+LH59	109	1984			✓
	LH82+LH74	104	1985	✓	LH74+LH51	110	1986	✓	✓	✓
	DK547	104	1986	✓	LH132+LH59	112	1986	✓	✓	✓
					LH132+LH109	115	1987			✓
1990	DK535	103	1988	✓	LH212+LH204	112	1989	✓	✓	
	DK512	101	1991	✓	DK614	111	1991	✓	✓	
	LH222+LH172	101	1991	✓	DK623	112	1991	✓	✓	
1995	DK527	102	1994	✓	DK626	112	1994			✓
	LH198+LH176	102	1995	✓	DK604	110	1994	✓	✓	✓
	DK546	104	1995	✓	RX730	114	1996	✓	✓	✓
2000	LH227+LH295	100	1998	✓	RX740	111	1998	✓	✓	
	DK539	103	1999	✓	DKC60–19	110	2001	✓	✓	
	DKC51–43	101	2002	✓	DKC60–08	110	2001	✓	✓	
2005					RX715	111	2004			✓
	DKC53–11	103	2004	✓	DKC63–78	113	2005			✓
	DKC52–21	102	2004	✓	DKC61–69	111	2005	✓	✓	✓
	DKC52–59	102	2006	✓	DKC63–42	113	2006	✓	✓	✓
					DKC61–21	111	2006	✓	✓	✓
2010	H5222VT3	102	2008	✓	DKC62–97	112	2011	✓	✓	
	DKC55–09	105	2009	✓	DKC61–88	111	2011	✓	✓	
	DKC53–56	103	2012	✓	DKC62–08	112	2012	✓	✓	
2015					DKC63–33	113	2013			✓
	DKC54–38	104	2014	✓	DKC60–67	110	2014			✓
	DKC55–20	105	2015	✓	DKC61–54	111	2015	✓	✓	✓
	DKC52–84	102	2015	✓	213–19VT2P	113	2016	✓	✓	✓
					DKC60–88	110	2017	✓	✓	✓
2020	DKC55–85	105	2018	✓	DKC63–57	113	2019	✓	✓	
	DKC53–25	103	2020	✓	213–93STX	113	2020	✓	✓	
	DKC52–16	102	2020	✓	DKC59–81	109	2020	✓	✓	

(Fig. 1) and included a large range of soil and weather conditions (Fig. S1). The topsoil texture (0–30 cm) was silty clay, silty clay loam, or clay loam (Table 1). The 0–30 cm soil organic matter ranged from 2.1% to 4.1%. Eleven of the thirteen locations had shallow water table at about 1.2 m from the surface, according to Soil Survey Staff (2021). Weather conditions were analyzed from May 1 (average emergence date) to September 5 (average end of grain filling date) using a gridded weather dataset (ERA5-Land, Muñoz-Sabater et al., 2021) provided by Bayer Crop Science. This dataset had hourly temperature, precipitation, relative humidity, and radiation data for a 30-yr period and proved accurate when compared to local station weather data available in some locations. Across environments, the cumulative in-season rain ranged from 263 to 550 mm (Fig. S1). Most western locations had 20% below normal precipitation, while for the rest, precipitation was near normal

(30-yr average). The cumulative solar radiation ranged from 2175 to 2413 MJ m^{-2}. Crops experienced average solar radiation except for Illinois locations, which experienced 4% below normal radiation, and for Blooming Prairie (MN), Dakota (NE), and Britt (IA), which experienced 4% above normal radiation (Fig. S1). The mean growing season temperature ranged from 20.8 to 23.6 °C, and the mean vapor pressure deficit ranged from 0.77 to 1.14 kPa across locations. Mean temperatures were 5% above the 30-yr average, and mean vapor pressure deficits were 20% above the 30-yr average except for Indiana and Illinois locations, which were near normal for both variables.

2.4. Plant measurements

At physiological maturity (R6; Abendroth et al., 2011), we measured aboveground plant biomass and its partitioning between stover and grain. Five consecutive plants were harvested from the center rows of each plot at the ground level and dried in a forced-air oven at 60 °C until constant weight. Grain and vegetative structures were weighted separately. The HI was computed by dividing the dry grain weight by the dry aboveground biomass weight. All hybrids were measured on the same date per experiment. Efforts were taken to select a date at which more than 95% of the hybrids had reached physiological maturity, a well-known challenge when working with hybrids that might differ slightly in their developmental progression. Plot level grain yield estimates were derived from combine harvesters (two middle rows per plot). Grain yield data are reported here at 150 g kg^{-1} moisture, while biomass data are reported at 0 g kg^{-1} moisture.

2.5. Statistical analysis

Initially, linear mixed-effects models were used to analyze whether the genetic gain of HI differed across environments and management practices. Models were adjusted to the four types of experiments, considering the year of hybrid release as a continuous variable and environment and management as factors. For 103-day and 111-day experiments, the relationship between the year of hybrid release and HI was initially investigated, including environment and year of release × environment interaction as random. For N-fertilization and plant density experiments, initial models considered the year of hybrid release as a fixed effect, and environment, management, and all possible interactions as random. The year of release × environment and the year of release × management interactions were dropped based on minimizing model Akaike's Information Criteria - AIC (Burnham and Anderson, 2004; Aho et al., 2014; Supplementary Table S1). Therefore, data from the different 111-day experiments were analyzed jointly. In the N-fertilization and plant density experiments, the low and sufficient N rates and the current and historical plant densities were considered in two different environments, resulting in fifteen environments for the analysis of the 111-day hybrids.

Next, we used random-effects models and ordinary least squares regressions to calculate the genetic gain of the HI in line with previous era studies (de la Vega et al., 2007; Haegele et al., 2013; Gizzi and Gambin, 2016; de Felipe et al., 2016). This approach provides a proper estimate of the genotype regardless of the effect of the growing conditions and unbalanced datasets (Smith et al., 2005; de la Vega et al., 2007; Gizzi and Gambin, 2016). First, the best linear unbiased predictors (BLUPs) for the HI of each hybrid were obtained from linear random-effects models using the function *lmer* in the *lme4* R package (Bates et al., 2015; R Core Team, 2021). The HI from each hybrid at each site was modeled as:

$$Y_{ijk} = \mu + \alpha_i + \beta_{j(i)} + \gamma_k + \alpha\gamma_{ik} + \varepsilon_{ijk} \quad (1)$$

where parameter Y_{ijk} is the HI from hybrid k, in the j block nested within the i environment; μ is the mean; α_i is the random effect of the i environment and is ~NID (0, σ^2_e); $\beta_{j(i)}$ is the random effect of j block nested within the i environment and is ~NID (0, σ^2_b); γ_k is the random effect of the k hybrid and is ~NID (0, σ^2_h); $\alpha\gamma_{ik}$ is the random effect of the interaction among i environment and k hybrid and is ~NID (0, σ^2_{he}); ε_{ijk} is the residual error and is ~NID (0, σ^2_ε). The BLUPs estimates for the model were obtained using the restricted maximum likelihood method. Then, the BLUPs obtained for each hybrid (γ_k) were plotted against the year of release, and ordinary least squares regressions were computed to estimate the HI genetic gain. We used the *SSlinp* function (*nlraa* package; Miguez, 2022) to fit the linear-plateau regression in 103-day hybrids and the *lm* function of R software to fit a linear regression in the 111-day hybrids. The genetic gain in HI was expressed both in absolute and relative terms per year. The relative gain was calculated as the absolute gain divided by the HI predicted in 2020 by the model (Fischer, 2015). Multiple comparisons between eras HI means were performed using the *LSD.test* function (*agricolae* package; de Mendiburu, 2021). We also performed a Pearson correlation analysis between locations' HI coefficients obtained from the models and soil properties, weather conditions, and management practices to determine which variables explained the differences in HI across environments.

2.6. Literature survey and analysis

We performed a literature review to compare our results with previous studies and to develop a general-use empirical maize HI model to describe the evolution of HI over a century of maize breeding. The literature search was conducted in Scopus on April 2022, using the keywords "maize" or "corn" or "Zea mays" together with "era" or "historical" or "decades", and "harvest index" or "HI" or "partitioning" or "allocation" published by 2021. We found 134 studies, but only ~10% fulfilled our criteria: data on HI or grain yield plus plant biomass to estimate HI from two or more maize hybrids released on different decades. Studies published in languages other than English and measurements from intercropping systems, inbred lines, open-pollinated products, or treatments with extreme management practices (i.e., zero N, > 12 plants m^{-2}) were excluded. Our final database comprised 16 peer-reviewed journal articles (Table 3) and our new 103-day and 111-day datasets. Eleven of these studies were from the United States, three from Argentina, three from China, and one from Canada. Most data were retrieved from tables, and few were digitized from figures when required.

The association between HI and hybrid's year of release was evaluated by fitting a non-linear mixed-effects model using *nlme* (*nlme* package; Pinheiro and Bates, 2000) and *SSplin* (*nlraa* package; Miguez, 2022) functions in R. The year of release was considered a fixed effect, while studies and experiments within a study (nested) were considered random effects. A single value per era was considered for studies where HI was reported as the average of different genotypes released in the same decade. Other models were fitted (linear, quadratic, and plateau-quadratic), but the plateau-linear was considered the best based on having the lowest Akaike's Information Criteria - AIC.

3. Results

3.1. Grain yield, biomass, and harvest index

Across environments and management treatments, the average grain yield for 111-day hybrids ranged from 9.8 to 15.9 Mg ha^{-1}, and for the 103-day hybrids, from 11.7 to 14.2 Mg ha^{-1} (Table 1). The dry biomass per plant was comparable between 103-day and 111-day hybrids and ranged from 226 g per plant (5th percentile) to 395 g per plant (95th percentile, Fig. S2). In the N-fertilizer experiments, the hybrids with sufficient N fertilizer had 49 g more biomass per plant than the hybrids with insufficient N fertilizer (Fig. S2). In the plant density experiments, the biomass per plant increased with the year of hybrid release at the current density treatment (8.1 plants m^{-2}) and decreased at the historical

Table 3
Description of the literature maize era studies including reference, country (US: United States of America; CAN: Canada; ARG: Argentina; CHI: China), hybrid's year of release, number of genotypes and environments (combinations of location-year) analyzed, proposed model equation by the study, and harvest index genetic gain (year^{-1}, linear gain) and slope significance (p-value) estimated with the reported values (Fig. S3). LP stands for linear-plateau.

N	Reference	Country	Years	Reported Genotypes	Number of Genotypes	Number of Environments	Model Fit	HI gain (year^{-1})	p-value
1	Tapper (1983)*	US	1930–1970	Public	20	4	Sigmoid	-0.0014	0.03
2	Meghji et al. (1984)*	US	1930–1970	Public	11	2	-	-0.0001	0.84
3	Russell (1985)*	US	1930–1980	Public	28	2	Linear	0.0006	0.15
4	Tollenaar (1989)	CAN	1959–1988	Various	8	4	-	0.0001	0.91
5	Luque et al. (2006)	ARG	1965–1997	Various	7	2	Linear	0.0047	< 0.01
6	Haegele et al. (2013)*	US	1967–2006	Dekalb	21	2	Linear	0.0016	< 0.01
7	Ma et al. (2014)	CHI	1967–2010	Public	9	2	Linear	0.0017	< 0.01
8	Chen et al. (2015)	US	1975–2005	Dekalb	4	2	Linear	0.0010	< 0.01
9	Zhao et al. (2015)	CHI	1978–2000	Public	4	2	Linear	0.0014	< 0.01
10	Chen et al. (2016)	US	1967–2005	Dekalb	8	4	Linear	0.0008	0.01
11	Di Matteo et al. (2016)	ARG	1965–2010	Dekalb	11	5	Linear	0.0024	0.01
12	Woli et al. (2016)*	US	1960–2000	Pioneer	10	2	Linear	0.0016	0.01
13	Emmett et al. (2018)	US	1936–2011	Pioneer	12	1	Linear	0.0008	0.01
14	Mueller et al. (2019)	US	1946–2015	Pioneer	7	2	Linear	0.0019	< 0.01
15	Curin et al. (2020)	ARG	1980–2012	Dekalb	4	5	LP	0.0011	< 0.01
16	Liu et al. (2021)	CHI	1982–2016	Public	6	2	Linear	0.0020	< 0.01
17	Bayer 103-day	US	1983–2020	Bayer	24	4	LP	0.0017	< 0.01
18	Bayer 111-day	US	1983–2020	Bayer	30	9	Linear	0.0013	< 0.01

* Harvest index reported as the average of genotypes released in the same decade.

density treatment (4.7–8.1 plants m^{-2}). However, at the unit area level (kg ha^{-1}), the biomass increased with the year of hybrid release in both plant density treatments (data not shown). The average HI was similar for the 103-day and 111-day hybrids (0.55 $\pm$ 0.04) and ranged from 0.471 to 0.643 (Table 4). The variability in HI was similar across eras, with an average coefficient of variation of 5.4% (Table 4). Lebanon (IN), on the treatment with low N had the lowest location-average HI (0.511), and Jerseyville (IL), on the treatment with sufficient N fertilizer had the highest HI (0.604).

3.2. Harvest index genetic gain in field experiments across US Corn Belt

The HI genetic gain was consistently positive across environments and management practices but differed between the relative maturity groups (Figs. 2 and 3). For the 111-day hybrids, the HI genetic gain was linear, increasing from 0.537 in 1983 to 0.584 in 2020, resulting in a relative gain of 0.22% per year (Fig. 2). For the 103-day hybrids, the HI genetic gain followed a linear-plateau relationship. The HI increased from 0.516 in 1983 to 0.571 in 2000, reaching a plateau (Fig. 3). The average relative genetic gain for the 37 years was 0.26% per year in the 103-day hybrids. Three 103-day hybrids (LH198+LH176, DK546, and LH227+LH295, Table 2) had very low HI values compared to coexisting hybrids and deviated from the overall pattern (see points in Fig. 3); these were excluded when adjusting the linear-plateau model. Coincidently, these three hybrids had high leaf area per plant and low reproductive partitioning during grain filling (unpublished results).

While the HI genetic gain was consistent among environments and management practices (Figs. 2 and 3), the average HI values differed across environments. The variability in planting date, N fertilizer, and soil water availability among environments partially explained this. Our data revealed a significant relationship between planting date and HI across environments – the earlier the planting, the higher the HI (Fig. 4). Hybrids receiving sufficient N fertilizer had 5% higher HI than hybrids with insufficient N fertilizer ($p < 0.001$; Fig. 2B, Fig. S2). At the rainfed Denton (NE) location, HI averaged 0.536, while at the irrigated Beatrice (NE) location, HI averaged 0.575. These two locations had similar environmental and management conditions (Table 1; Fig. S1), except for the water availability. The Denton (NE) location received 263 mm (rainfall) while the Beatrice (NE) location received 404 mm (315 mm from rainfall and 89 mm from irrigation). Plant density in the studied levels (current vs. historical) did not explain variability in HI within an era (Fig. 2B, Fig. S2). Lastly, we did not find a significant association between location average HI and soil properties (i.e., soil organic matter, texture), weather conditions, or management practices other than planting date.

3.3. Synthesis of current findings with literature studies

Published era studies in the 1980s reported non-significant or slightly negative changes in the HI (studies #1–4), but recent studies reported positive genetic gains (studies #5–18 in Table 3 and Fig. S3). The magnitude of the genetic gain varied 5-fold between studies. The response of HI to the year of hybrid release was linear in most cases. The combined analysis (18 datasets) revealed a plateau-linear relationship between HI and the year of hybrid release (Fig. 5). There was no genetic gain in HI from 1930 to 1964, but a significant linear HI increase from 0.473 in 1964 to 0.554 in 2020. The multi-dataset average HI relative genetic gain was 0.26% per year.

Table 4
Harvest index summary statistics per era for 103 and 111-day Bayer hybrids. 5th percentile, mean, 95th percentile, and coefficient of variation (CV).

Maturity	Statistic	1985	1990	1995	2000	2005	2010	2015	2020
103-day	5th percentile	0.487	0.510	0.471	0.508	0.544	0.547	0.542	0.542
	Mean*	0.513 c	0.543 b	0.522 c	0.549 b	0.576 a	0.575 a	0.567 a	0.570 a
	95th percentile	0.539	0.576	0.589	0.596	0.598	0.602	0.597	0.606
	CV (%)	3.8	4.1	7.5	6.1	3.0	3.3	3.4	3.8
111-day	5th percentile	0.490	0.485	0.506	0.489	0.514	0.501	0.526	0.532
	Mean*	0.541 e	0.547 de	0.544 e	0.554 cd	0.573 b	0.562 c	0.579 a	0.586 a
	95th percentile	0.595	0.619	0.590	0.624	0.617	0.622	0.622	0.643
	CV (%)	5.8	7.9	4.6	8.2	6.0	7.0	5.1	6.1

* Different letters indicate significant differences at $p < 0.05$.

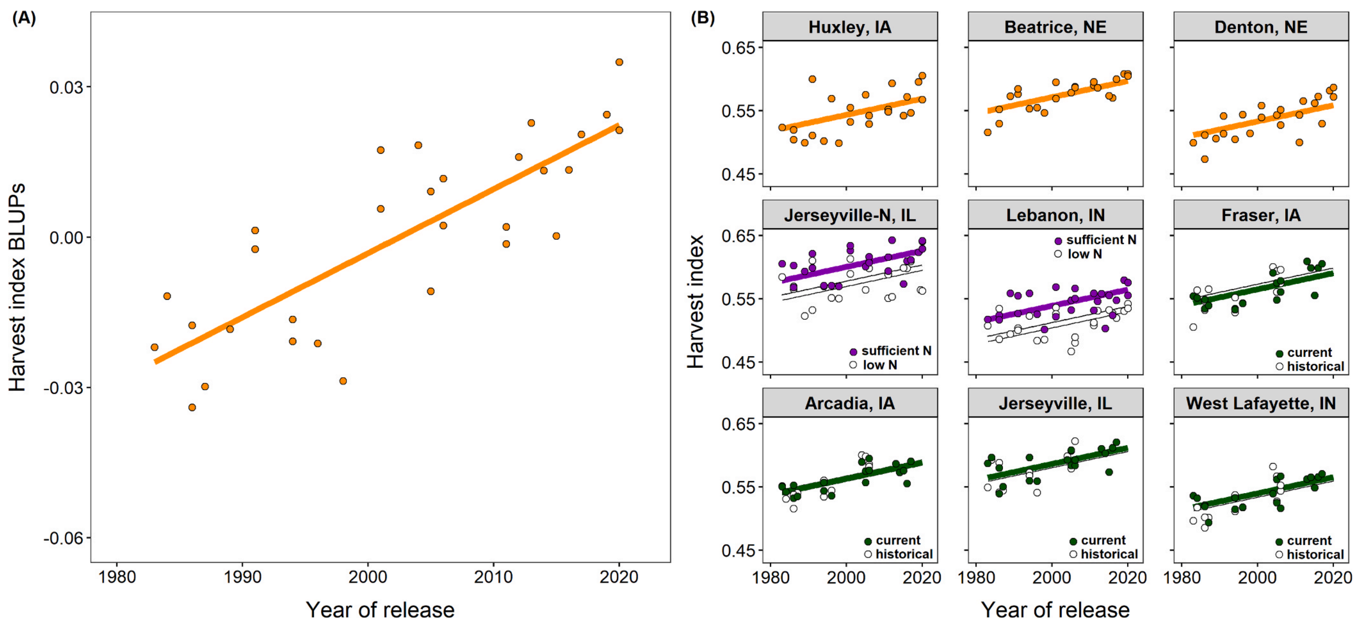

Fig. 2. : Harvest index genetic gain in 111-day hybrids. In panel A, the symbols represent HI's best linear unbiased predictors (BLUPs), while the line represents the ordinary least square regression model (y = −2.560 + 0.0013 *x, R^2 = 0.65). In panel B, the symbols represent the observed HI for the hybrids in each experiment (including management factors), while the lines represent the genetic gain model from panel A. The equations for each location are listed in the Suppl. Table S2. In G×E×M plant density experiments, historical plant density was 4.7, 5.9, 7.0, and 8.1 plants m^{-2} for hybrids released in eras 1985, 1995, 2005, and 2015, respectively, while the current density treatment was 8.1 plants m^{-2} for all eras.

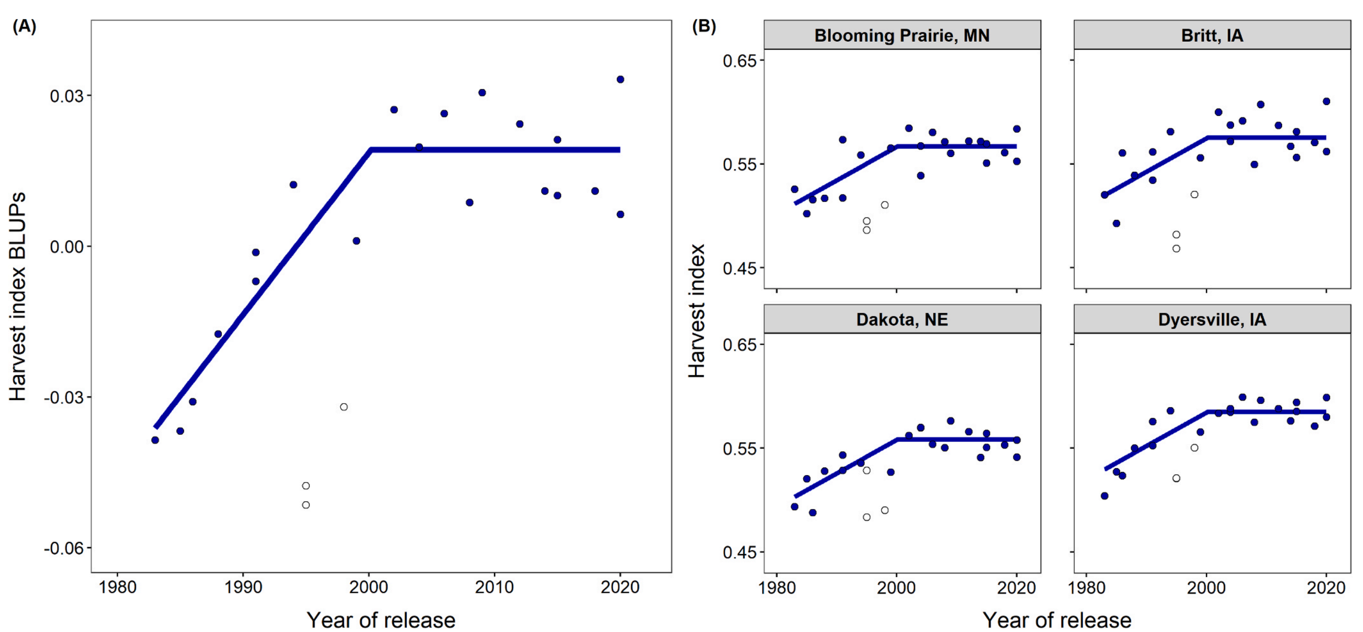

Fig. 3. : Harvest index genetic gain in 103-day hybrids. In panel A, the symbols represent HI best linear unbiased predictors (BLUPs), while the line represents the ordinary least square regression model (y = −6.429 + 0.0032 *x if x < 2000, and y = 0.019 if x ≥ 2000, R^2 = 0.84). In panel B, the symbols represent the observed HI for the hybrids tested in each experiment, while the lines represent the genetic gain model from panel A. The equations for each location are listed in the Suppl. Table S2. Open symbols indicate hybrids that were excluded from the regression model.

4. Discussion

Our study is unique in terms of the scale of experimentation (13 locations, 54 hybrids, and plant density and N-fertilizer treatments) and is the first to report the HI genetic gain for Bayer Crop Science Legacy maize hybrids (not only Dekalb brand, Table 2). Furthermore, we developed a robust HI model (Fig. 5) by synthesizing new findings with previously published articles to support future research. We showed that maize HI has increased by 17% over the last 56 years, which rectifies the misconception that maize HI has remained unchanged by breeding.

4.1. The historical increase in HI is attributed to maize breeding, not to management

The estimated genetic gain for the Bayer hybrids was similar across environments, plant density, and N-fertilizer treatments, suggesting that maize breeding is the main cause for the increase in HI (Figs. 2 and 3). Different environments and N-fertilizer rates consistently altered the

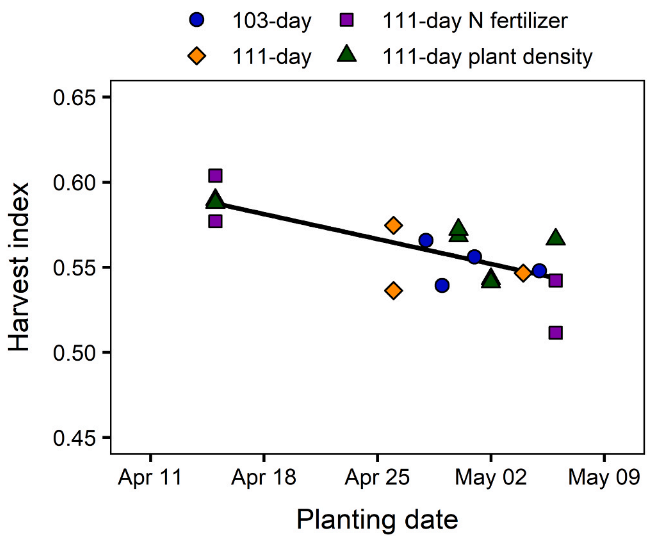

Fig. 4. : The relationship between the experimental average HI and planting date (y = 0.617–0.002 *x, where x is days after April 1; R^2 = 0.49).

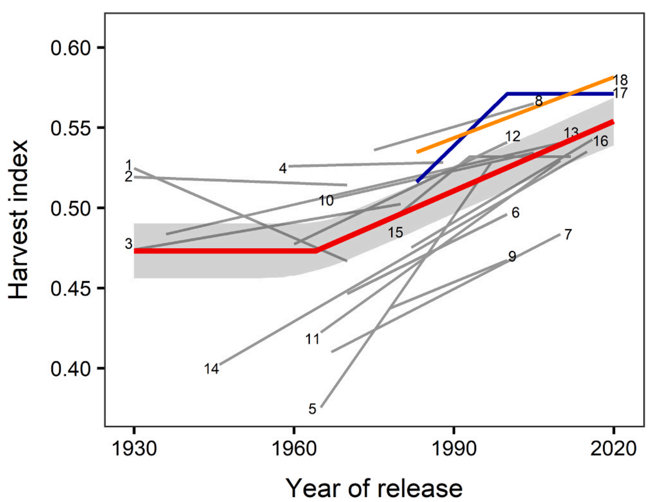

Fig. 5. : The evolution of maize harvest index over a century of breeding. Numbered grey lines refer to literature era studies (Table 3), and the blue and orange lines to data from this study (Table 3, Figs. 2–3). The red line represents the multi-dataset overall model adjusted with mixed models, considering study and experiment within study as random, shaded areas represent the 95% confidence interval from the model. The multi-dataset equation is y = 0.473 if x < 1964, and y = 0.473 + 0.0014 * (x - 1964) if x ≥ 1964.

magnitude of the HI across all hybrids. A higher plant N uptake can increase the HI (Sinclair, 1998), which is confirmed by our results as the sufficient N-fertilization level increased the HI by ~ 5% consistently across all hybrids (Fig. 2B). Haegele et al. (2013), Chen et al., (2015, 2016), and Woli et al. (2016) tested hybrids released in different years with different N levels and reported that the interaction between the year of hybrid release and N level on HI was not significant, which is consistent with our results. In our study, experiments were planted on different dates (Table 1), and a significant relationship between HI and planting date was identified: the earlier the planting date, the higher the HI (Fig. 4). Our findings agree with Bonelli et al. (2016), who attributed the decrease in HI associated with later planting to the higher reduction in dry matter accumulation during the post-silking vs. pre-silking period. From an agronomic perspective, trends toward earlier planting dates in the US Corn Belt (1.3 days per decade; Kucharik, 2006; Baum et al., 2020) might have slightly contributed to the HI increase over the years.

Vega et al. (2000) and Echarte and Andrade (2003) reported that maize HI is slightly reduced when the individual plant biomass is > 480 g plant^{-1} and severely reduced when it is < 150 g plant^{-1}. In our study, although the variability in biomass per plant was large (216–478 g plant^{-1}; Fig. S2), no extreme high or low values were recorded to reduce the HI due to the plant density. In fact, planting old hybrids at high density (8.1 plants m^{-2}) did not reduce the HI (Fig. 2).

Our study revealed that the HI genetic gain differs among hybrids' relative maturity. The genetic gain increased linearly in the 111-day hybrids (Fig. 2) while it reached a plateau in the 103-day hybrids (Fig. 3). This may be explained by the fact that the genetic gain analyses are time-sensitive and 103-day hybrids might have encountered a temporal plateau as the genetic gain is mostly driven by breakthrough lines that do not show up every year (e.g., 111-day hybrids released in 2015 and 2020 had major gains compared to previous eras, Table 4). Also, there might be a lower physiological limit for 103-day than 111-day hybrids HI, which could be explained by the reduced growing window in northern environments that imposes a limit to the grain fill period duration. Considering that HI is a compound trait that results from the interplay of several physiological processes, future studies should explore the underlying causes explaining HI genetic gain through field experimentation and simulation crop modeling.

In the literature studies, the HI genetic gain was linear in most cases, seeming to plateau in only two studies that used 120-day hybrids (Di Matteo et al., 2016; Curin et al., 2020). A plateau in HI does not mean that grain yields are plateauing. Di Matteo et al. (2016) and Curin et al. (2020) showed yield increases despite HI plateauing. Different traits (e.g., biomass production) have increased more to compensate for the small gain in HI. In our study, 103-day and 111-day experiments showed a linear increase in grain yield with year of hybrid release (unpublished data). Studies published in the 1980s reported non-significant or slightly negative changes in the HI (Table 3, Fig. S3), generating the misconception that there was no genetic gain for HI. Studies published in the last decade, together with our results, provide strong evidence for a positive genetic gain in HI (Table 3, Fig. S3, Fig. 5).

4.2. The choice of hybrids can influence the genetic gain; the use of one hybrid per decade is not recommended

Our results indicate that contemporary hybrids within an era exhibit considerable variability in HI (Fig. 5) and that this variability is similar for old and new hybrids (Table 4). For example, in the 110-day hybrids of the 2000 era, the HI ranged from 0.489 to 0.624 (Table 4). This has two implications. First, there is genetic variability to further increase this trait through breeding programs. Second, the estimated genetic gain can significantly change depending on the selection of hybrids, especially when only one hybrid per decade is considered. Therefore, results based on era studies evaluating one genotype per decade can have large uncertainties (Woli et al., 2016). Robust studies with many hybrids across environments, such as the present study, are needed to accurately estimate the genetic gain of a trait. An alternative option to accurately estimate the true genetic gain of a plant trait is to analyze it across multiple era studies (Fig. 5).

4.3. Implications of the harvest index increase for productivity and sustainability assessments

Our synthesis of literature with current findings (Fig. 5, Table 3) revealed that there was no genetic gain in HI from 1930 to 1964 but a significant linear HI increase since 1964 (Fig. 5). The developed model is robust as it accounts for many different environments and genotypes. This model has important implications for more accurate estimations of soil carbon inputs, for supporting crop modeling studies, and for enhancing our understanding of the breeding effect on grain yield gain. For example, Messina et al. (2022) estimated that the increase in radiation use efficiency is responsible for 32% of the historical yield increase in the US Corn Belt but assumed that the HI remained constant over the century. Our study revealed that the HI has increased, suggesting that the contribution of the radiation use efficiency on the historical yield increase has probably been overestimated. This example demonstrates the importance of studying secondary traits to correctly estimate relative contributions to grain yields.

The HI is a critical parameter in soil carbon budget studies as it converts grain yields to stover (carbon) inputs. Many previous studies have used a fixed HI value over the years (Prince et al., 2001; Huang et al., 2007), while others increased HI values over the years (Johnson et al., 2006; Tan et al., 2012). The developed HI evolution model (Fig. 5), which accounts for genotypes from different breeding programs and diverse environments across the world (Table 3), can facilitate more robust estimations of carbon and nutrient budgets (Ludemann et al., 2022). To illustrate this importance, we estimated that in Johnson et al.'s (2006) study that spanned 60 years (1940–2000), using the HI values from our study (Fig. 5) would have resulted in 42% less carbon input in 1940 and 3% more carbon input in 2000 compared to the reported values.

Our results can inform various crop model applications, including decision support and climate change assessments. The HI is a key input in some models, such as SWAT (Neitsch et al., 2011), and an emergent property (output) in models such as APSIM (Keating et al., 2003) and DSSAT (Jones et al., 2003). Regardless of input or output, having baseline HI information for over a century (Fig. 5) is of great importance for two reasons. First, the models can be better trained/tested to reflect the historical yield increase (and thus better capture soil nitrogen cycling) instead of using post-simulation yield detrending methods (Baum et al., 2020). Secondly, HI scenarios can be generated from Fig. 5 to drive future grain yield predictions. Currently, crop models ignore maize breeding genetic gains in future climate yield predictions (Rosenzweig et al., 2014; Baum et al., 2020). We hypothesize that the inclusion of genetic gain will challenge existing estimates of climate change impacts on crop production. In this context, more era studies on secondary traits such as leaf appearance rate, grain fill duration, or nitrogen use efficiency (e.g., DeBruin et al., 2017; Mueller et al., 2019, dos Santos et al., 2022), are needed to inform crop modeling.

4.4. Harvest index accounts for 15% of the historical yield increase in the US Corn Belt

In the US Corn Belt, maize grain yields increased from 5.2 Mg ha^{-1} in 1970 to 11.1 Mg ha^{-1} in 2020 (114% increase; Table 5; USDA-NASS, 2022b). This increase incorporates all contributing factors such as genetics, management, and environment. Given that yield is the product of biomass and HI, using USDA NASS yields and the reported increase in HI (15% from 1970 to 2020; Fig. 5), we estimate that biomass has increased by 86% from 1970 to 2020 (Table 5). We can then determine that the HI genetic gain over the last 50 years of maize breeding in the US accounted for ca. 15% [= 15% / (15% + 86%)] of the historical maize yield increase in the US Corn Belt. Our finding adds new knowledge on the causes of the historical maize yield increase in the US Corn Belt (Duvick, 2005; Tollenaar et al., 2017; Messina et al., 2021, 2022; Rizzo et al., 2022). For reference, Rizzo et al. (2022) attributed 13% of the historical maize yield increase to breeding, Messina et al. (2022) attributed 32% of the increase to radiation use efficiency, Tollenaar et al. (2017) attributed 27% of the increase to brightening increasing radiation, while Duvick (2005) attributed 50% of the increase to breeding and 50% to management.

Table 5
Grain yield, biomass, and harvest index gains from 1970 to 2020 for the US Corn Belt. Grain yield was obtained from USDA-NASS data, while biomass considering the harvest index proposed model (Fig. 5) and USDA-NASS grain yield data. Grain yield is reported at 150 g kg^{-1} moisture, while biomass is reported at 0 g kg^{-1}.

	Grain yield	Crop Biomass	Harvest index
Year	Mg ha^{-1}	Mg ha^{-1}	-
1970	5.21	9.19	0.482
2020	11.15	17.11	0.554
Absolute gain	5.94	7.92	0.072
Relative gain since 1970	114%	86%	15%

4.5. Perspectives for future genetic improvement

Our synthesis-analysis revealed a 0.0014 year^{-1} increase in HI since 1964, the beginning of the green revolution (Fig. 5). If this trend continues, assuming crop improvement continues at the same rate, the HI will be 0.042 higher in 2050 compared to today's levels (ca. 0.55–0.60). Because there are physiological limits to HI, it may prove challenging to increase HI beyond an upper limit. Currently, the upper limit is unknown, and estimation of it is rather complex because of the strong coupling between HI and crop nitrogen uptake (Sinclair, 1998) as well as the tradeoffs between biomass production and tissue N concentrations (Ciampitti et al., 2021). In general, the higher the crop nitrogen uptake, and/or the lower the tissue nitrogen concentration (either grain or stover), the higher the HI (Sinclair, 1998; Ciampitti and Vyn, 2012; Mueller et al., 2019). Further improvements in maize reproductive growth (decreased barrenness, increased kernel numbers or weight, extended grain fill period, and increased photosynthesis) will likely increase the HI. However, we could expect a natural maximum imposed by the minimum structure biomass required to sustain the grain at maturity.

Compared to other cereal crops such as wheat and rice, in which breeding has increased the HI by 30% over the last century (Hay, 1995; Hutsch and Schubert, 2017), the increase in maize HI is less (17%; Figs. 2, 3, and 5). This may suggest that there is room for further improvements. In fact, recent advances in genetics with the development of the short stature maize hybrids that have higher HI compared to the isogenic tall stature maize hybrids, illustrates that further increases in HI are very likely to occur (Paciorek et al., 2022). Furthermore, most studies examined here indicated strong linear increases in HI over recent decades (Fig. 5), supporting that HI is likely to increase further in the coming years. Such an increase will benefit grain yield and food security.

5. Conclusions

This study provides the most comprehensive assessment of maize HI evolution over years of commercial maize breeding programs and different agronomic management practices such as plant density and nitrogen fertilizer. We provide a model that describes the temporal evolution of maize HI over a century, which can support future estimations of stover carbon inputs in sustainability studies and inform crop models to better capture historical yield increases. Our new experimental data from 13 environments, together with 16 literature studies, showed a 0.26% year^{-1} relative genetic gain in HI since 1964. This increase in HI accounted for 15% of the historical maize yield increase in the US. We attributed the historical increase in HI to maize breeding, not management. Planting hybrids from the 80s and 90s at the current plant density did not reduce the HI, and the N-fertilizer effect on HI was similar between hybrids from different eras. Future era studies could evaluate more extreme plant densities, N-fertilizer levels, and environments to further enhance our understanding of management effects on the HI.

CRediT authorship contribution statement

AR: Conceptualization, Formal analysis, Writing – original draft; **SVA**: Conceptualization, Funding acquisition, Supervision, Project administration, Writing – review & editing. **AR, KK, MB, CdS, GJND, CSS, EFE, GK, AT,** and **LOP**: Investigation, Writing – review & editing; **KRL, MG, MC, ST, PSS, CT,** and **TJV**: Funding acquisition, Writing – review & editing. All authors contributed to the manuscript.

Declaration of Competing Interest

ST, DME, and MG are employed by Bayer Crop Science. PSS is a co-lead of the Genomes to Fields Initiative and PI of the USDA-NIFA funded Agricultural Genome to Phenome Initiative. He is co-founder of Data2-Bio, LLC; Dryland Genetics, Inc.; EnGeniousAg, LLC; and LookAhead Breeding, LLC. He is a member of the scientific advisory board and a shareholder of Hi-Fidelity Genetics, Inc., and a member of the scientific advisory boards of Kemin Industries and Centro de Tecnologia Canavieira. He is a recipient of research funding from Iowa Corn and Bayer Crop Science. The remaining authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Data Availability

Data will be made available on request.

Acknowledgments

This study was supported by FFAR (project title: Evaluating the relative influence of maize breeding, field management, and environmental setting on crop production and sustainability targets), Bayer Crop Science, Leopold Center for Sustainable Development, the Plant Science Institute of Iowa State University, the Fulbright Program, and the Argentina Ministry of Education. We thank Bayer's maize breeding testing team for setting up and managing the field experiments. We also thank Garrett Verhagen from Purdue University, Emily Wright and Mickala Stallman from Iowa State University for their assistance with data collection, Brenda Gambín from Iowa State University for her constructive suggestions to the manuscript, Clarice Mensah from Bayer Crop Science for coordinating field experiments, and Bobby Brauer from Bayer Crop Science for providing the weather data.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.fcr.2023.108991.

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